

COLUMBIA UNIVERSITY

LAMONT-DOHERTY EARTH OBSERVATORY
THE EARTH INSTITUTE AT COLUMBIA UNIVERSITY

Volcanology **I**nfrastructure for **C**omputational **T**ools and **R**esources or VICTOR, is a cloud-based cyberinfrastructure platform for volcanologists

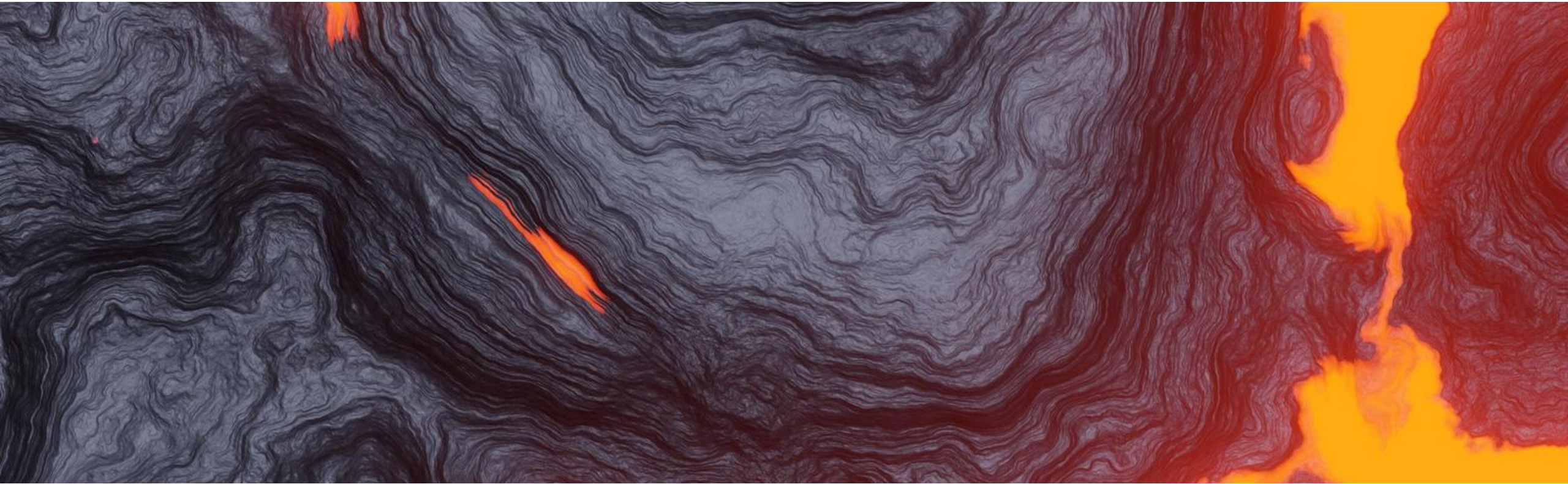


VICTOR is the successor to Vhub. However, the software and associated infrastructure are aging and inflexible, and a new medium was required

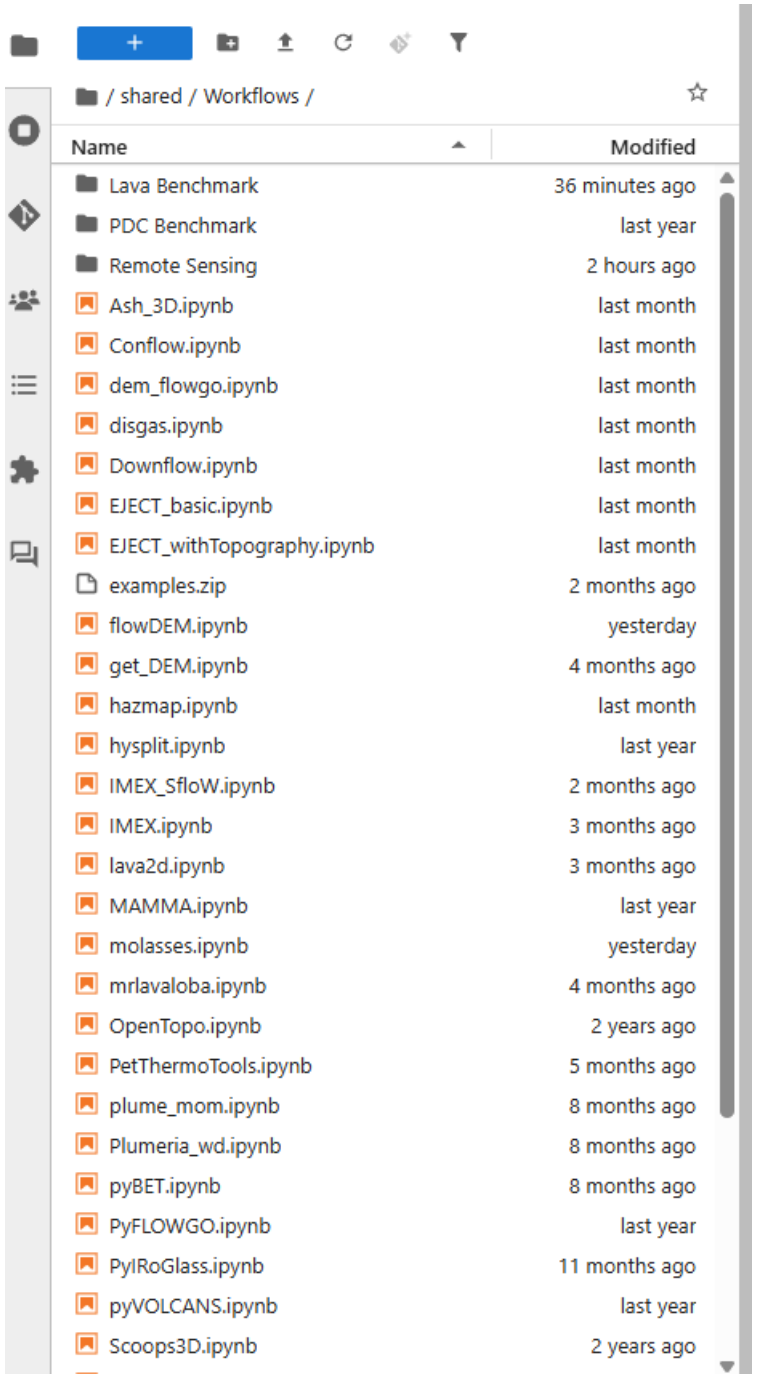
VICTOR

- VICTOR is meant to be a platform for every step of the computational process.
- Here, you can write, run, and analyze models and their associated data.
- Software libraries connect you to every corner of the cloud.

- VICTOR will help volcanologists at universities and observatories to find, use, compare, and link models of eruption processes by providing a central access portal and standardized workflows based on a JupyterHub environment.



Volcanic process/topic	Models
Tephra and plumes	Ash3d, Fall3D, HySPLIT, Plumeria, Tephra2, PLUME-MoM-TSM, Puffin, Hazmap
Gas dispersal, degassing	Disgas, sulfur_X, twodee, VolFe, VESical
PDC, granular flows	TITAN2D, VolcFlow, IMEX_SfloW2D
Lahars	LAHARZ, VolcFlow, IMEX_SfloW2D
Lava flows	Lava2D, MOLASSES, MrLavaLoba, Q-Lavha, Flowy, VENUSS, IMEX-Lava, PyFLOWGO, VolcFlow
Ballistics	EJECT
Flank collapse/failure	Scoops3D, clawpack
Magma thermodynamics/conduit processes/petrology	alphaMELTS, Conflow, MAGEmin, MAMMA, Perple_X, pyflowgo, Diadfit, pyIRoGlass, PyMELT, Thermombar, CONFORT
Multi-hazard, Bayesian Event Tree	pyBET, PyVOLCANS, Hazmap
Deformation tools	Magmaxisym, GNSSsphereTopoP, dMODELS



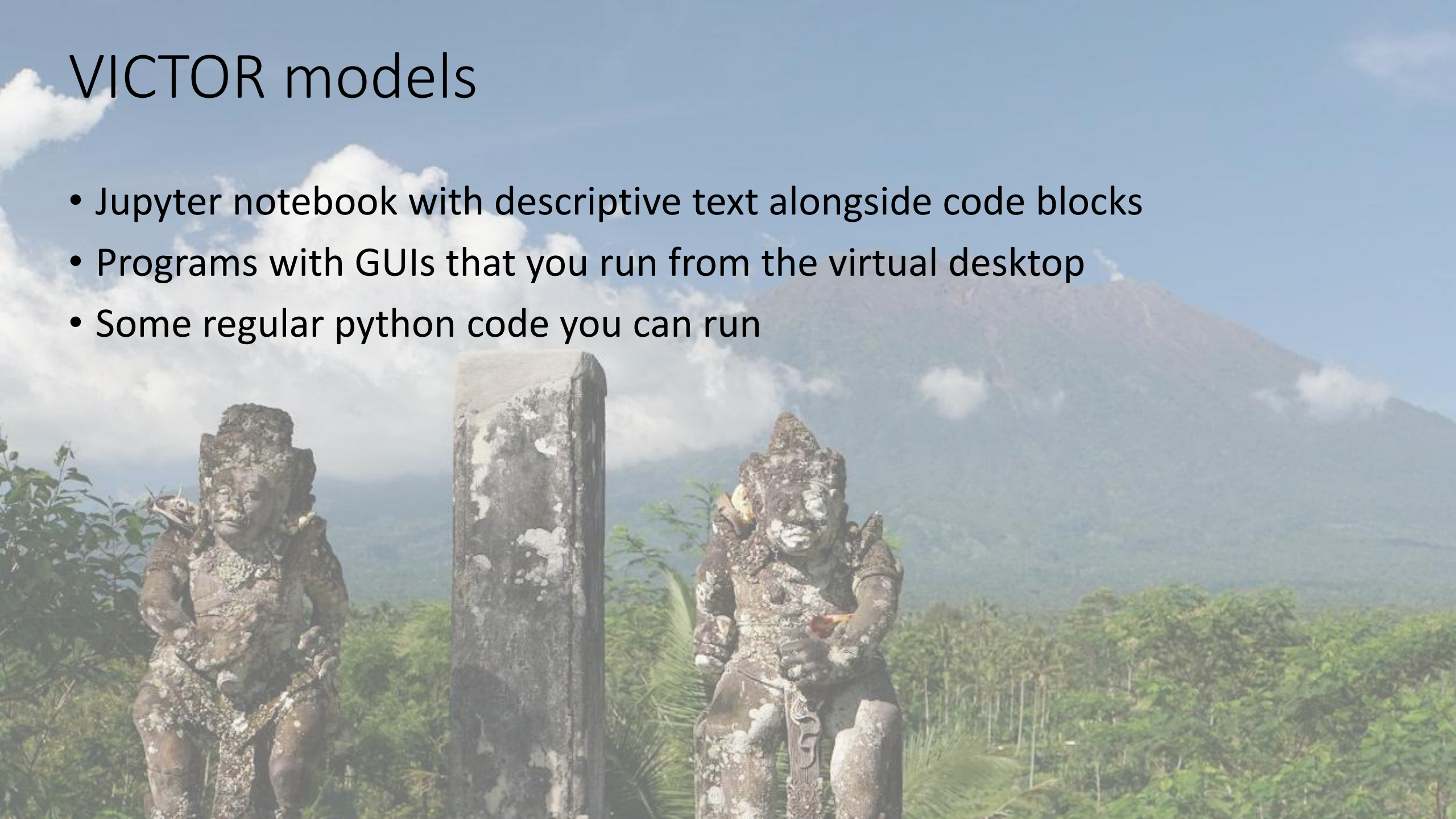

```
Activating profile: /srv/conda/etc/profile.d/conda.sh

When reporting any research conducted on VICTOR, please cite both our paper (https:/,

(notebook) jovyan@jupyter-sogburn:~$ victor setup
Hello, and welcome to the VICTOR platform.
We have a huge variety of models and tools for use.
This program is meant to make it simple to set up workflows and associated files.
To set up, please choose the number next to the model listed below.
1: alphaMELTS
2: ASH3d
3: CONFORT15
4: DISGAS
5: EJECT
6: flowDEM
7: Flowy
8: HAZMAP
9: HYSPLIT
10: IMEX (Lava)
11: IMEX_SfloW2D_V2
12: LaharZ
13: LAVA2D
14: MR LAVA LOBA
15: MOLASSES
16: PLUME-MoM-TSM
17: pyBET
18: PYFLOWGO/DOWNFLOWGO
19: PyIRoGlass
20: Scoops3D
21: SULFUR_X
22: TEPHRA2
23: TITAN2D
24: TWODEE
Model:
```

VICTOR models

- Jupyter notebook with descriptive text alongside code blocks
- Programs with GUIs that you run from the virtual desktop
- Some regular python code you can run



Jupyter notebooks contain both the code for running the model and text that describes what the code is doing, what parameters you need to change, and explanations about the parameter choice

Vital general parameters. [Change these]

In the cell below, choose the format of the DEM and specify its path. Then, provide a scale for the simulation to accommodate, as well as the gravity of the planet being simul
interations are another optional input. Choosing the frequency of outputs by either iteration number and/or simulator time is vital to having comprehensive time series analys

[26]:

```
# You can choose GIS_GRASS or GDAL.
# For the CSAV exercise choose 'GDAL'
gis_format='GDAL'

# Path to the DEM.
# For the CSAV exercise, choose the DEM we reprojected. Always make sure the file is in UTM!
# You can right click on the DEM in your files and choose "copy file path", but make sure the file below starts with "/home/jovyan/"
dem="/home/jovyan/Titan2D/Licancabur_utm_small_dem.tif"

#Length scale for simulation (m).
# For the CSAV exercise, we know Licancabur can produce PDCs up to 15 km, so change this to 20000 m.
length_scale=20000.0

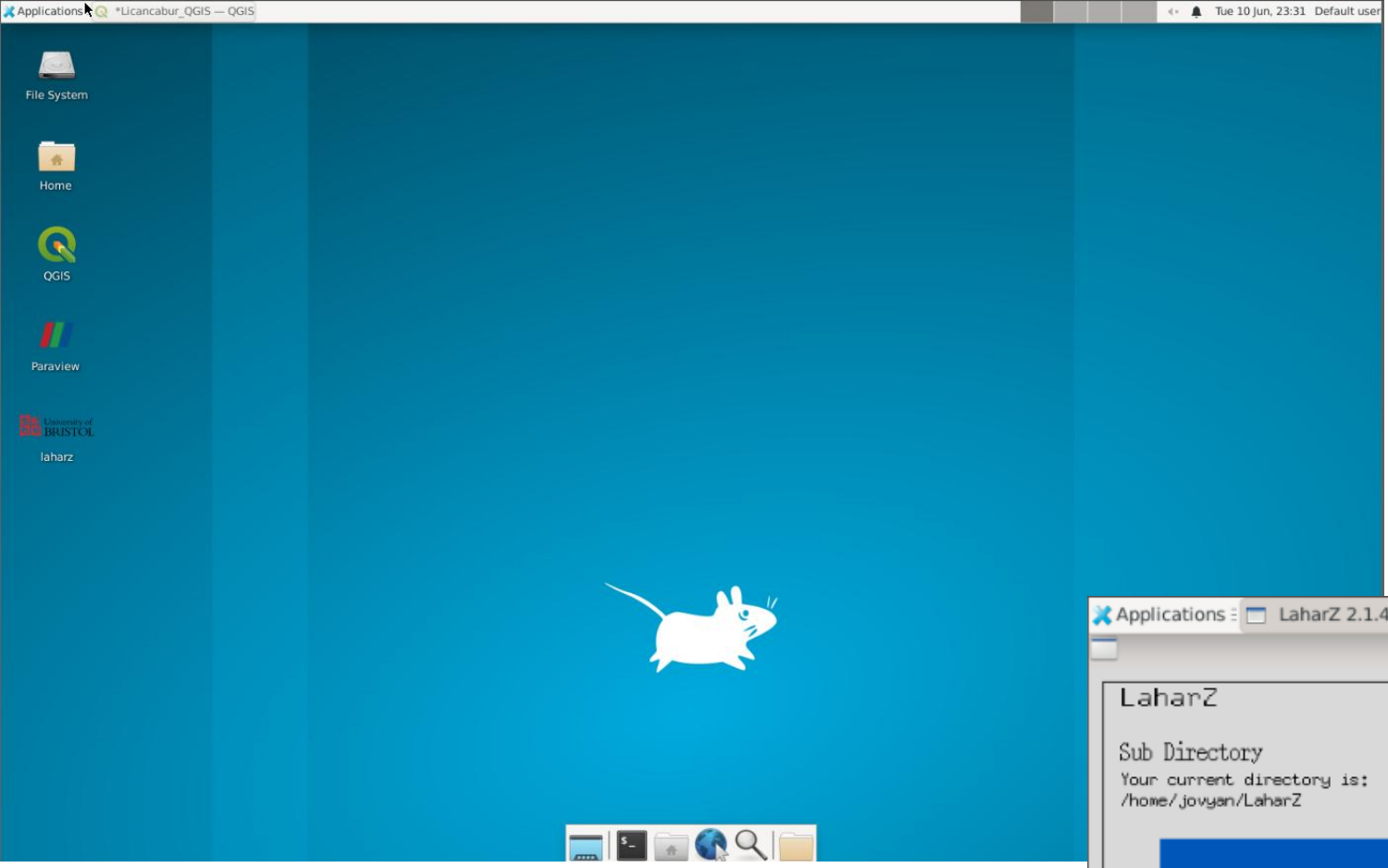
#Gravity for simulation. Don't change unless your volcano is on another planet.
gravity=9.8

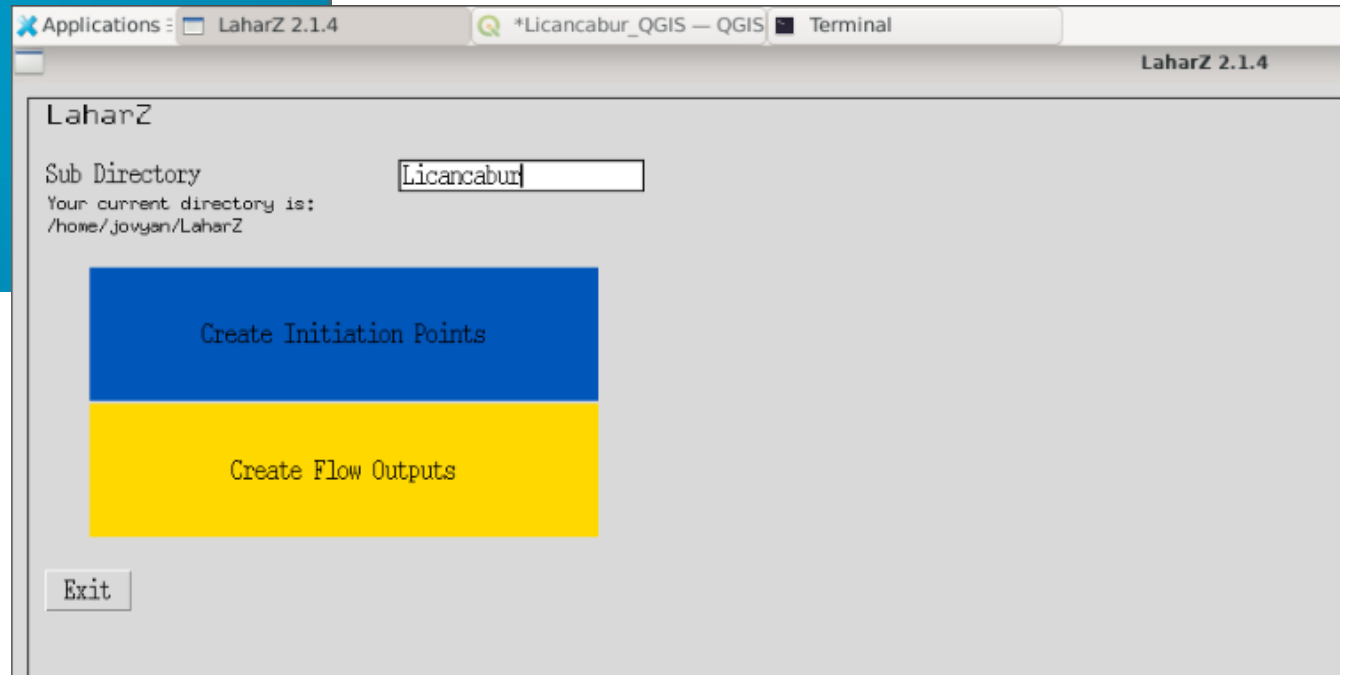
# Set maximum iterations of the simulation (max_iter) and the maximum simulation time (max_time).
# For the CSAV exercise, let's use a max_iter of 100000 and simulate 10 minutes (600 sec) of time by entering '600' for 'max_time'
max_iter=100000
max_time=600

#Set frequency and format of outputs.
#choose from {'xdmf', 'tecplot', 'meshplot','grasssites'}
#We recommend xdmf for use with paraview

#For the CSAV exercise we will leave this as "xdmf"
vizoutput="xdmf"

#Set interval for model outputs, either number of iterations (diter) or number of simulation seconds (dtime), both are valid.
#For the CSAV exercise, we want this to run relatively fast, so we will only make an output every 60 sec. Set 'dtime' = '60'
diter=None
dtime=60
```


Applications
*Licancabur_QGIS — QGIS
Tue 10 Jun, 23:31
Default user
File System
Home
QGIS
Paraview
laharz


Applications
LaharZ 2.1.4
*Licancabur_QGIS — QGIS
Terminal
LaharZ 2.1.4
LaharZ
Sub Directory
Licancabur
Your current directory is:
/home/jovyan/LaharZ
Create Initiation Points
Create Flow Outputs
Exit

More resources about VICTOR

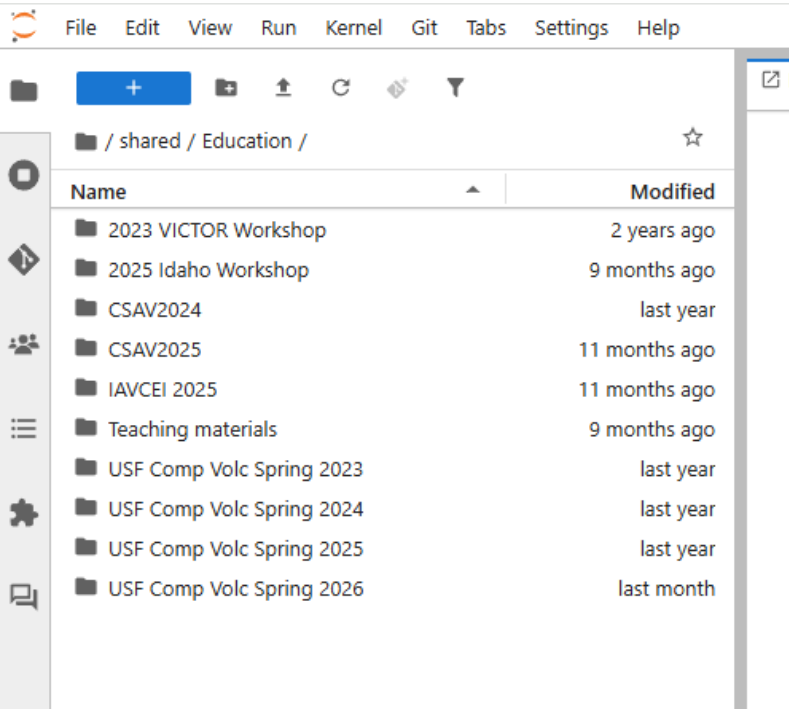
- https://victor.ldeo.columbia.edu/
- https://victor.ldeo.columbia.edu/content/education
- https://victor.ldeo.columbia.edu/content/documentation
- https://www.youtube.com/watch?v=3wgfqFVSYwk

Documentation

For complete VICTOR documentation, use the Read The Docs. Or, use the table below to navigate to documentation specific to each model.

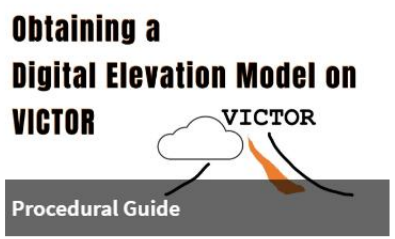
Models

Volcanic Process	
Lava Flows	IMEX-Lava, pyFLOWGO, MrLavaLoba, MOLASSES, VolcFlow, VENUSS, Lava2d
PDCs	TITAN2D, IMEX-SfloW2d
Particulate Dispersal	Ash3D, HAZMAP, HySPLIT, Fall3D, Tephra2, PLUME-MoM-TSM, Plumeria_wd
Gas dispersal	DISGAS, TWODEE

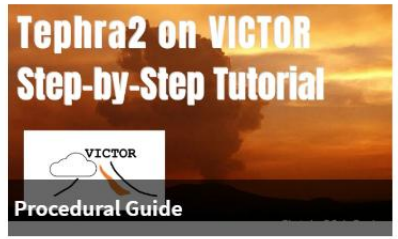


Topic Specific Tutorial Library

Clicking on the thumbnails below will open the resource in a new tab.

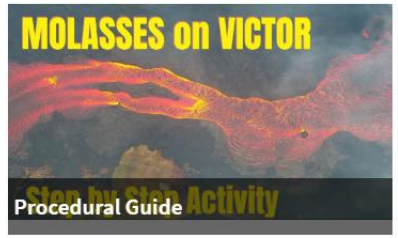












Walk-through Exercise: VICTOR set-up and finding course files

- https://hub.victorproject.org/



Operated by: **2i2c** | Funded by: **National Science Foundation** | Designed by: **2i2c**

Log in to continue

Selected Identity Provider

GitHub

Rem ction

Type to

GitHub

Google

Microsoft

By logging on to th the privacy policy.

Server Options

CPU Only

Image: Victor Notebook

Resource Allocation: ~4 GB RAM, ~0.5 CPUs (Recommended for Lighter Models/Scripting)
Up to ~4 CPUs when available

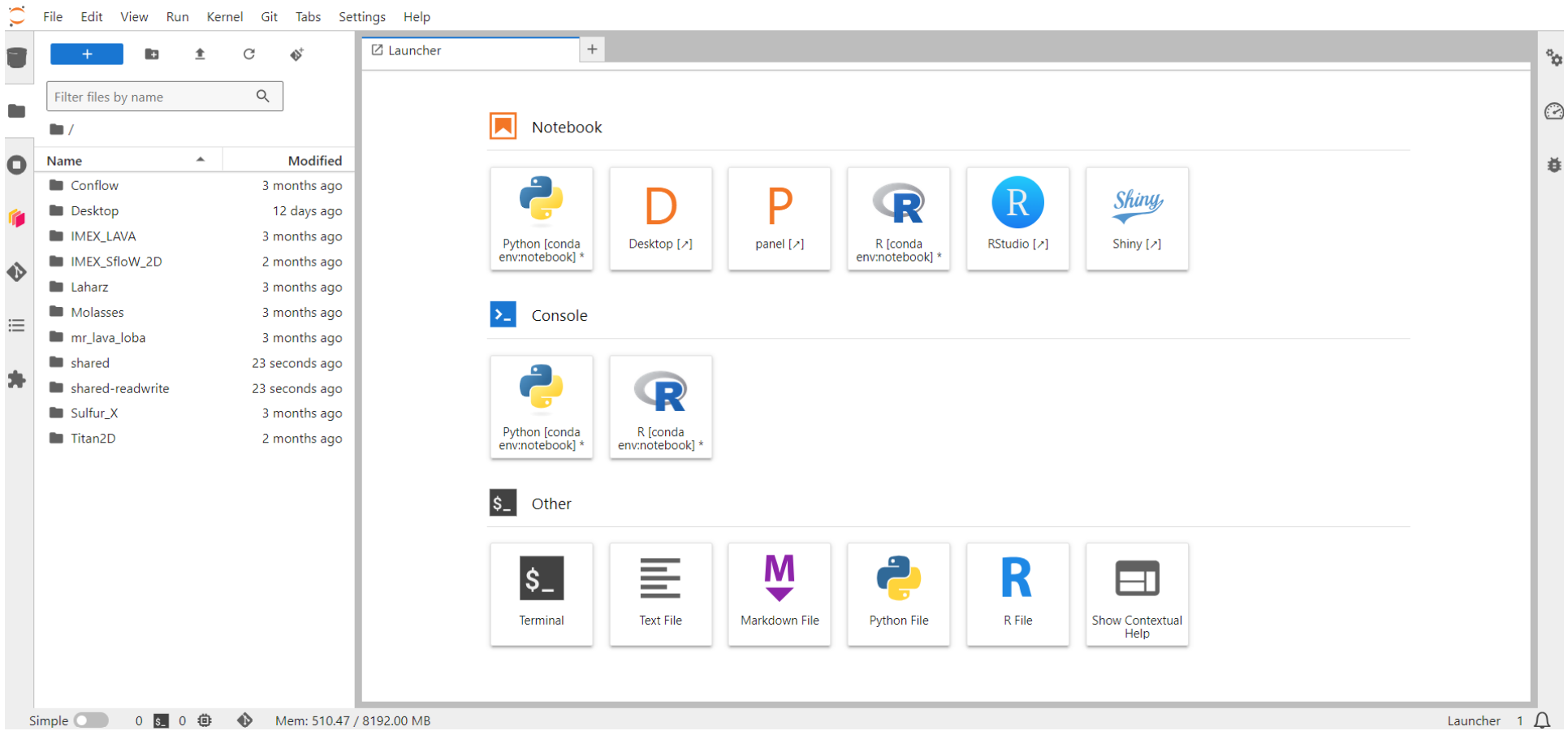
- ~4 GB RAM, ~0.5 CPUs (Recommended for Lighter Models/Scripting)
  Up to ~4 CPUs when available
- ~7 GB RAM, ~0.9 CPUs (Recommended for Heavier Models/Larger Datasets)
  Up to ~4 CPUs when available
- ~15 GB RAM, ~1.8 CPUs
  Up to ~4 CPUs when available
- ~29 GB RAM, ~4 CPUs (Recommended for Some Remote Sensing Workflows)

NVIDIA Tesl

Start a container on a dec

Image

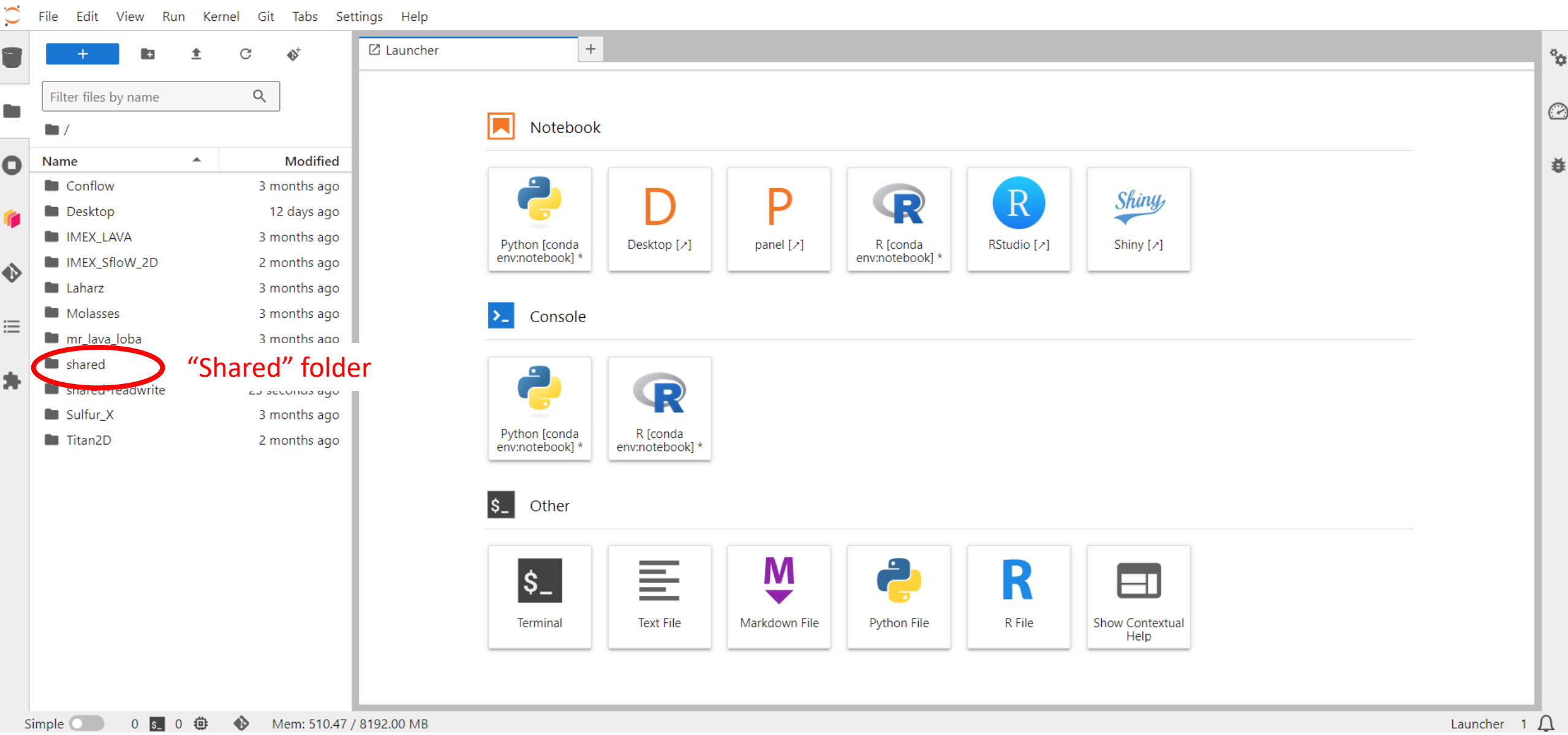
Start

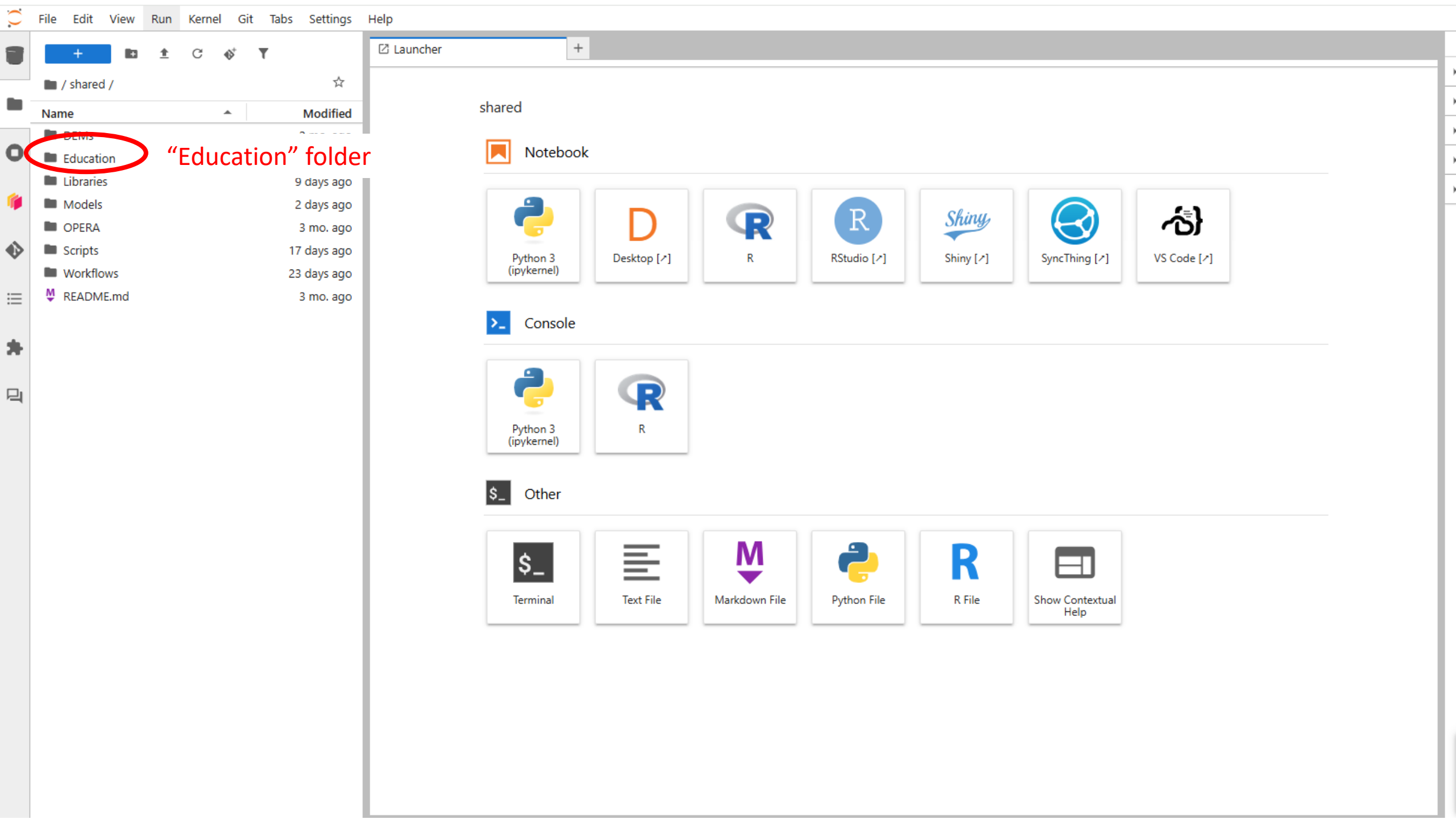


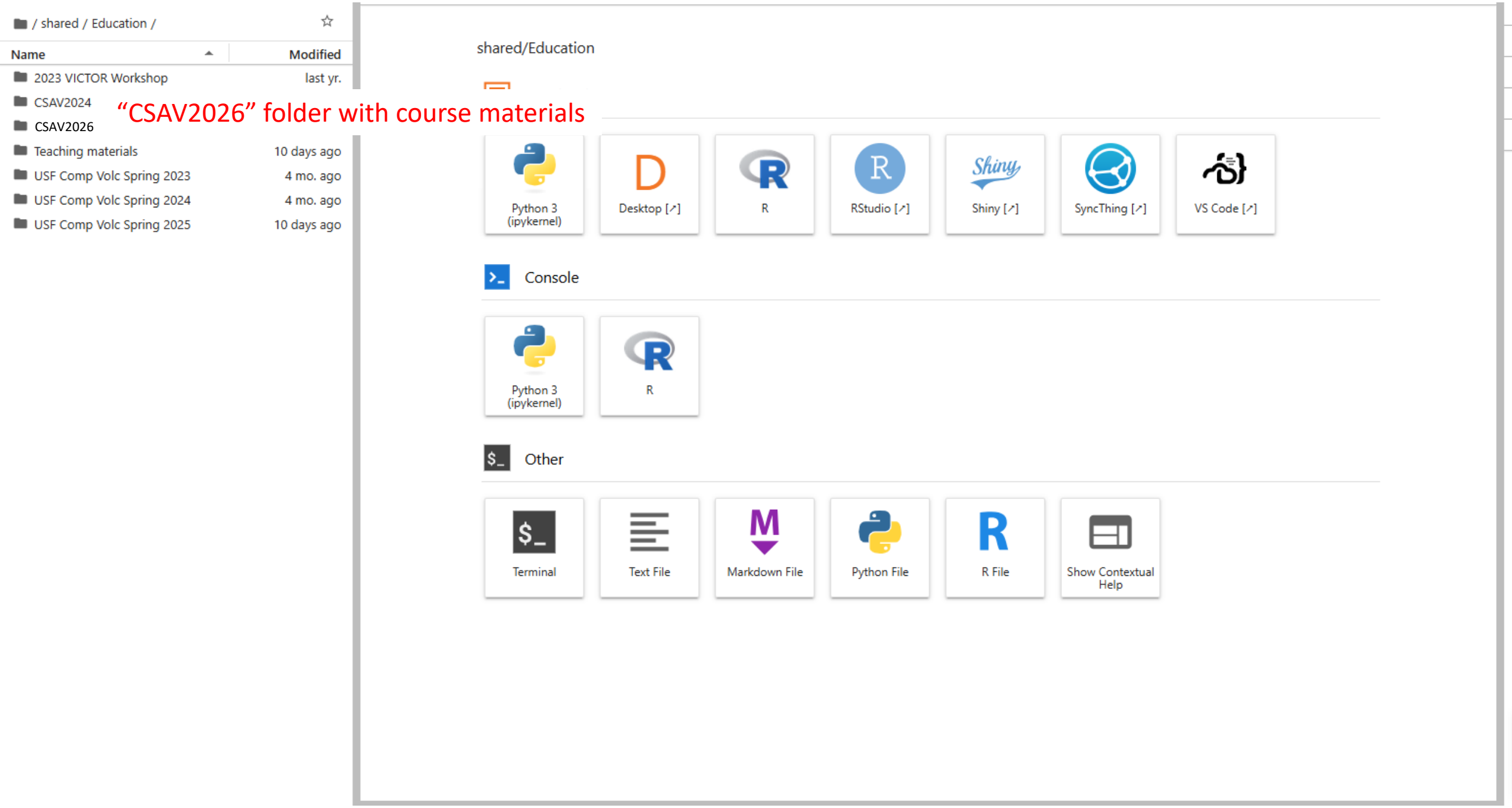
- When you login to VICTOR, your screen will look something like this
- Your files are on the left panel (but will look different than mine)
- Your “Launcher” has useful icons

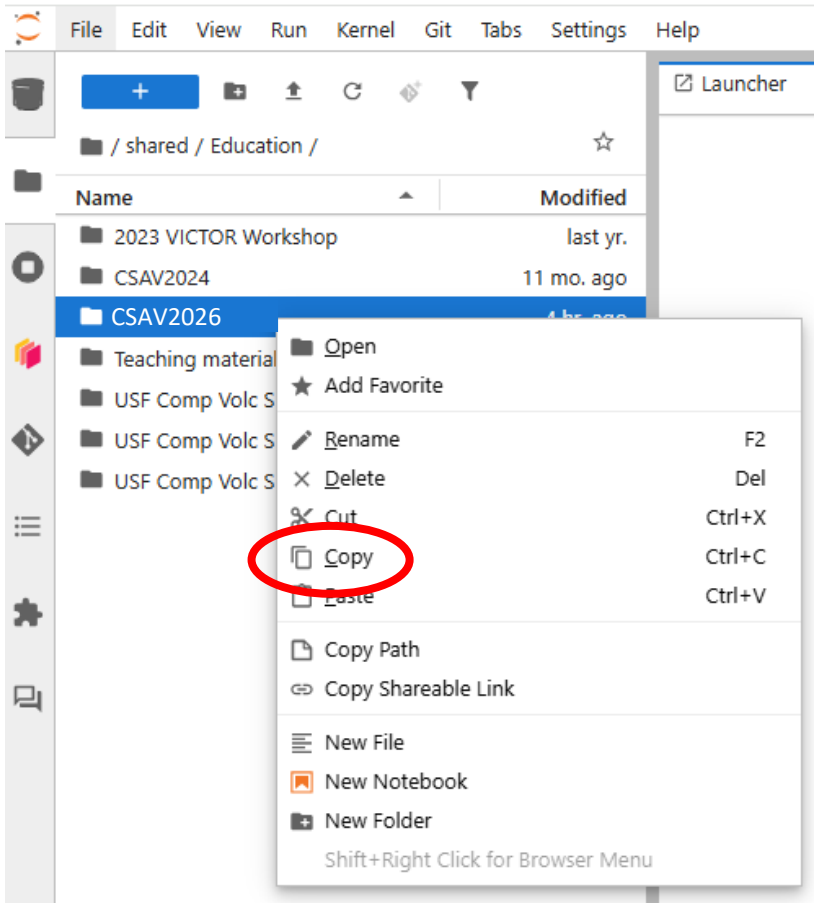
Tip: If you lose your “Launcher” screen, click the “plus” icon at the top left for “New Launcher”

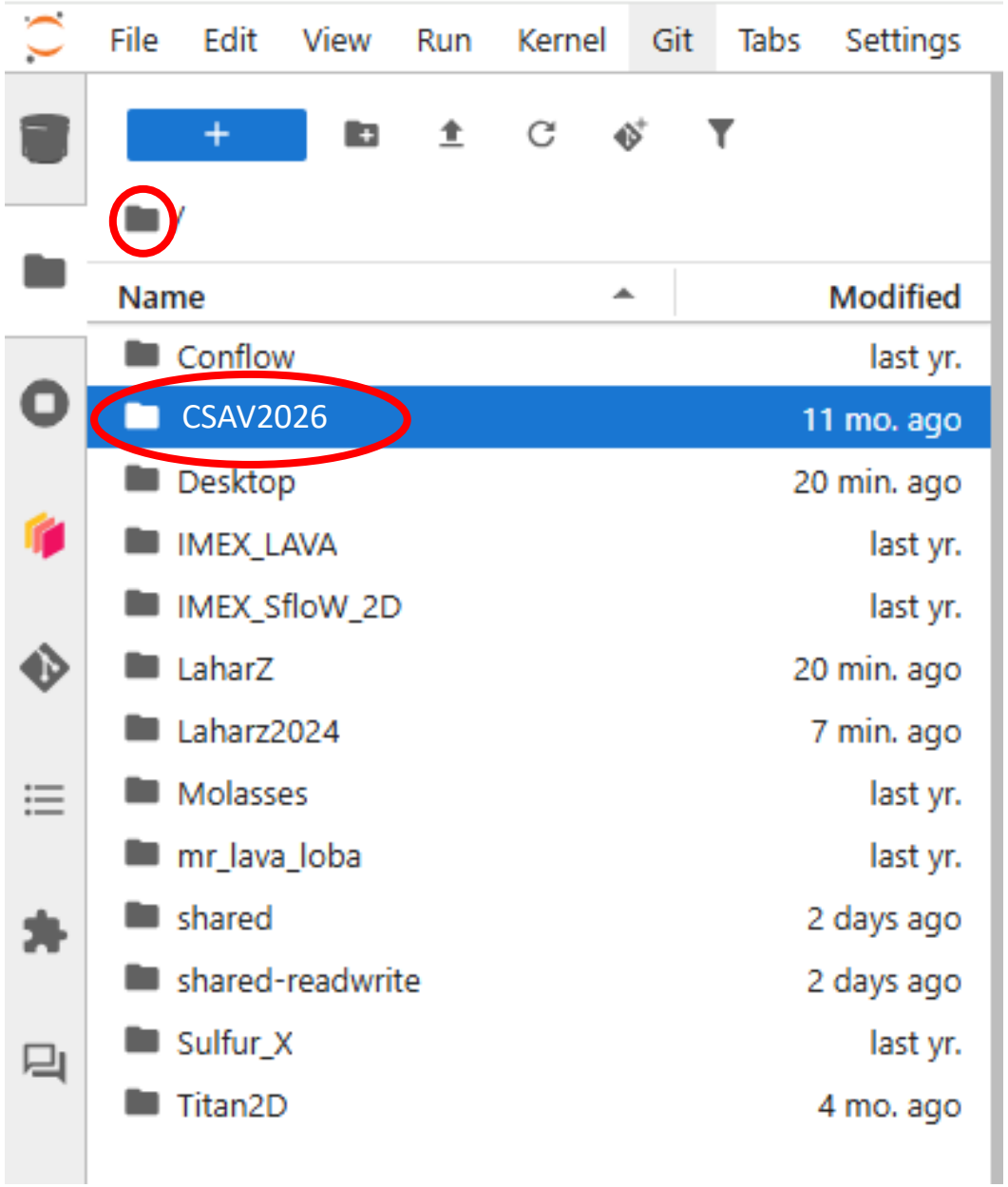
Our course files are in /shared/Education/CSAV2026



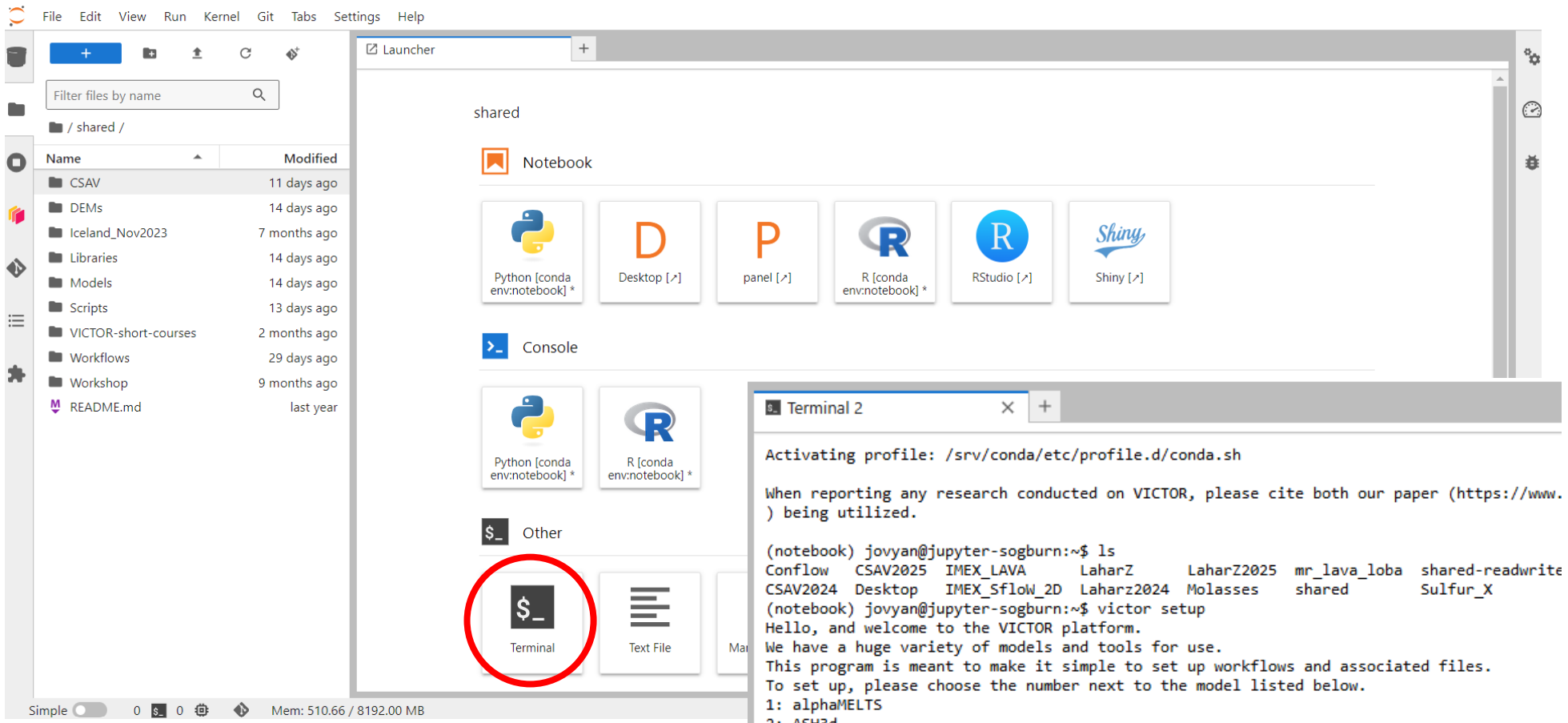

File
Edit
View
Run
Kernel
Git
Tabs
Settings
Help
/ shared /
Name
Modified
Education
Libraries
9 days ago
Models
2 days ago
OPERA
3 mo. ago
Scripts
17 days ago
Workflows
23 days ago
README.md
3 mo. ago
“Education” folder
Launcher
shared
Notebook
Python 3 (ipykernel)
Desktop [↗]
R
RStudio [↗]
Shiny [↗]
SyncThing [↗]
VS Code [↗]
Console
Python 3 (ipykernel)
R
Other
Terminal
Text File
Markdown File
Python File
R File
Show Contextual Help

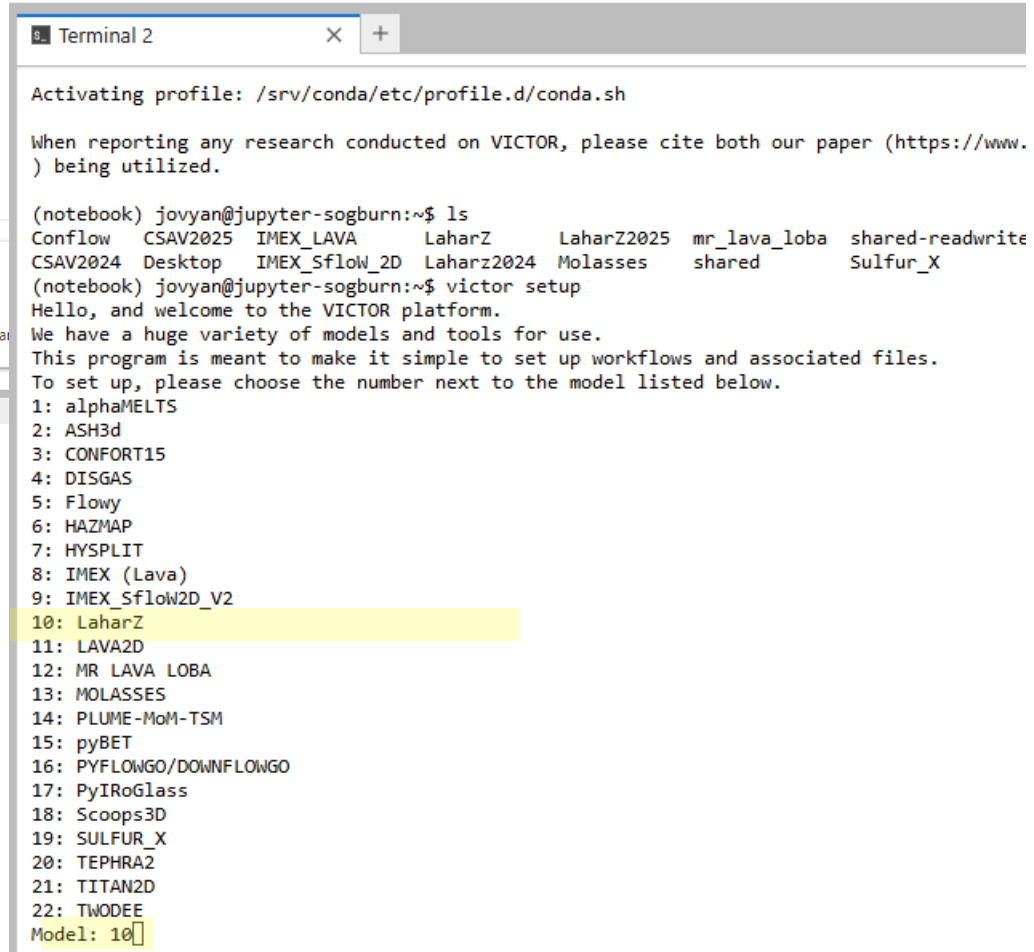

/ shared / Education /
Name
Modified
2023 VICTOR Workshop
last yr.
CSAV2024
CSAV2026
Teaching materials
10 days ago
USF Comp Volc Spring 2023
4 mo. ago
USF Comp Volc Spring 2024
4 mo. ago
USF Comp Volc Spring 2025
10 days ago
"CSAV2026" folder with course materials
shared/Education
Python 3 (ipykernel)
Desktop [↗]
R
RStudio [↗]
Shiny [↗]
SyncThing [↗]
VS Code [↗]
Console
Python 3 (ipykernel)
R
Other
Terminal
Text File
Markdown File
Python File
R File
Show Contextual Help





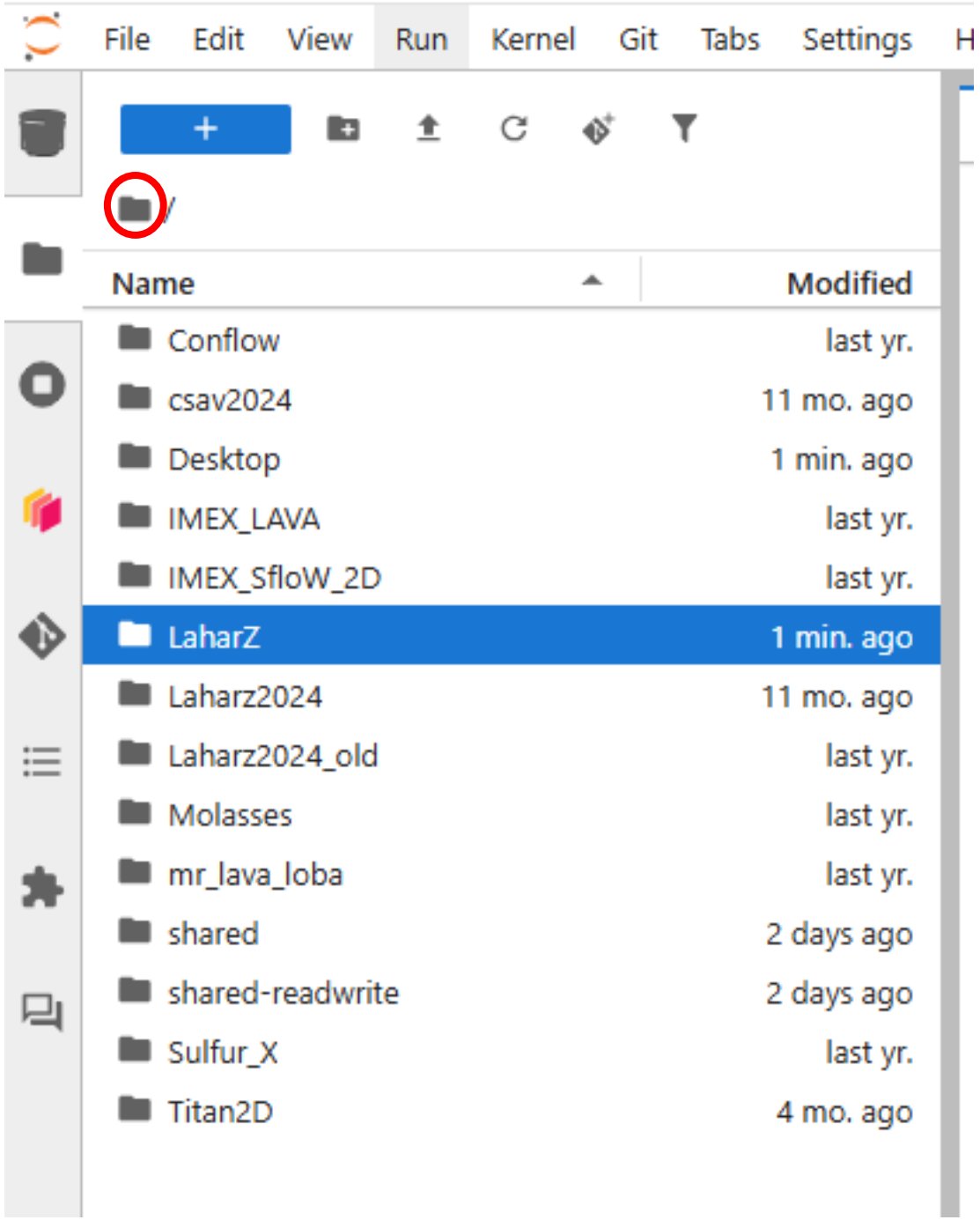
- Right click on the “CSAV2026” folder
- Click “Copy”
- Navigate to your home folder (click the folder icon)
- Paste the “CSAV2026” folder there
- (You will have different files than me, don’t worry!)

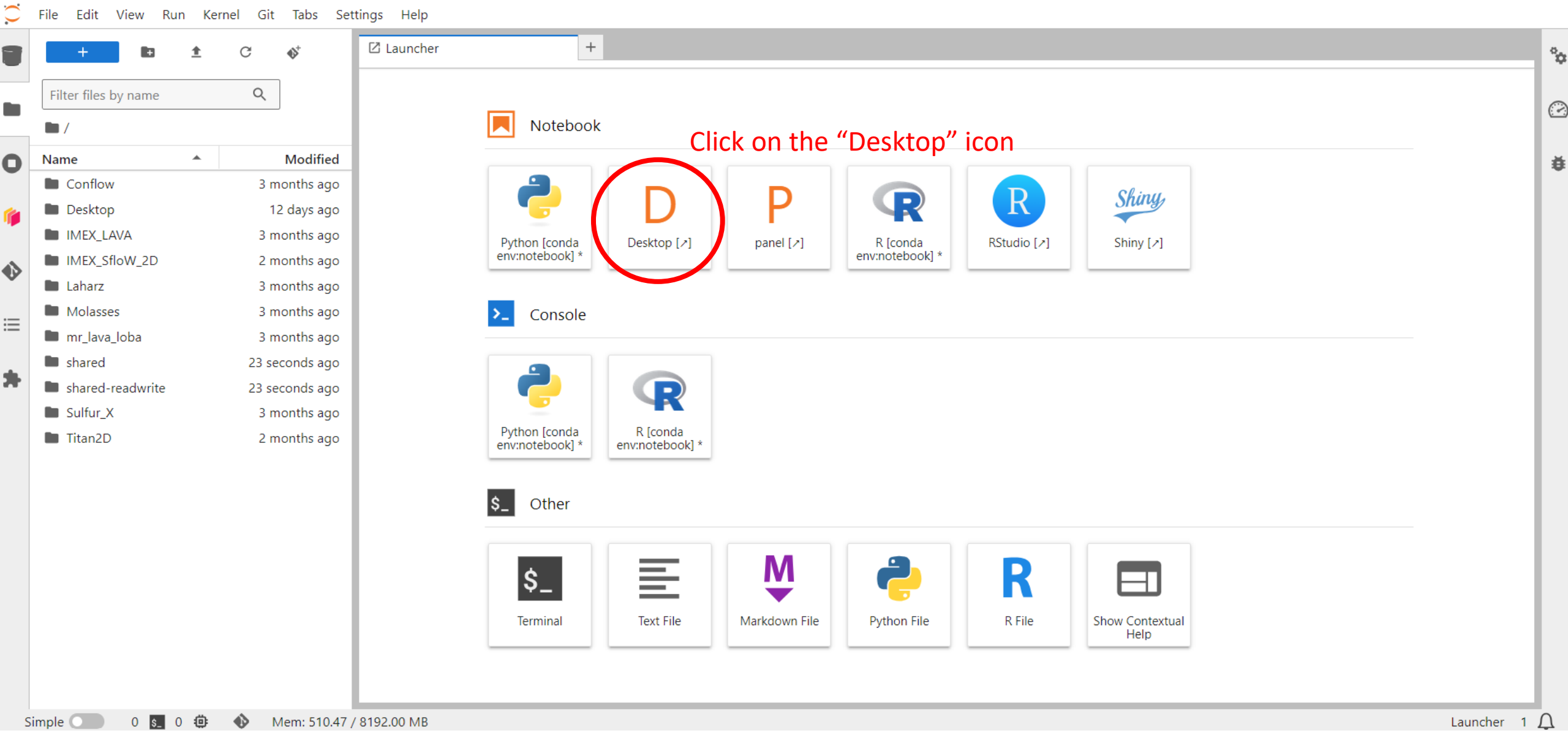


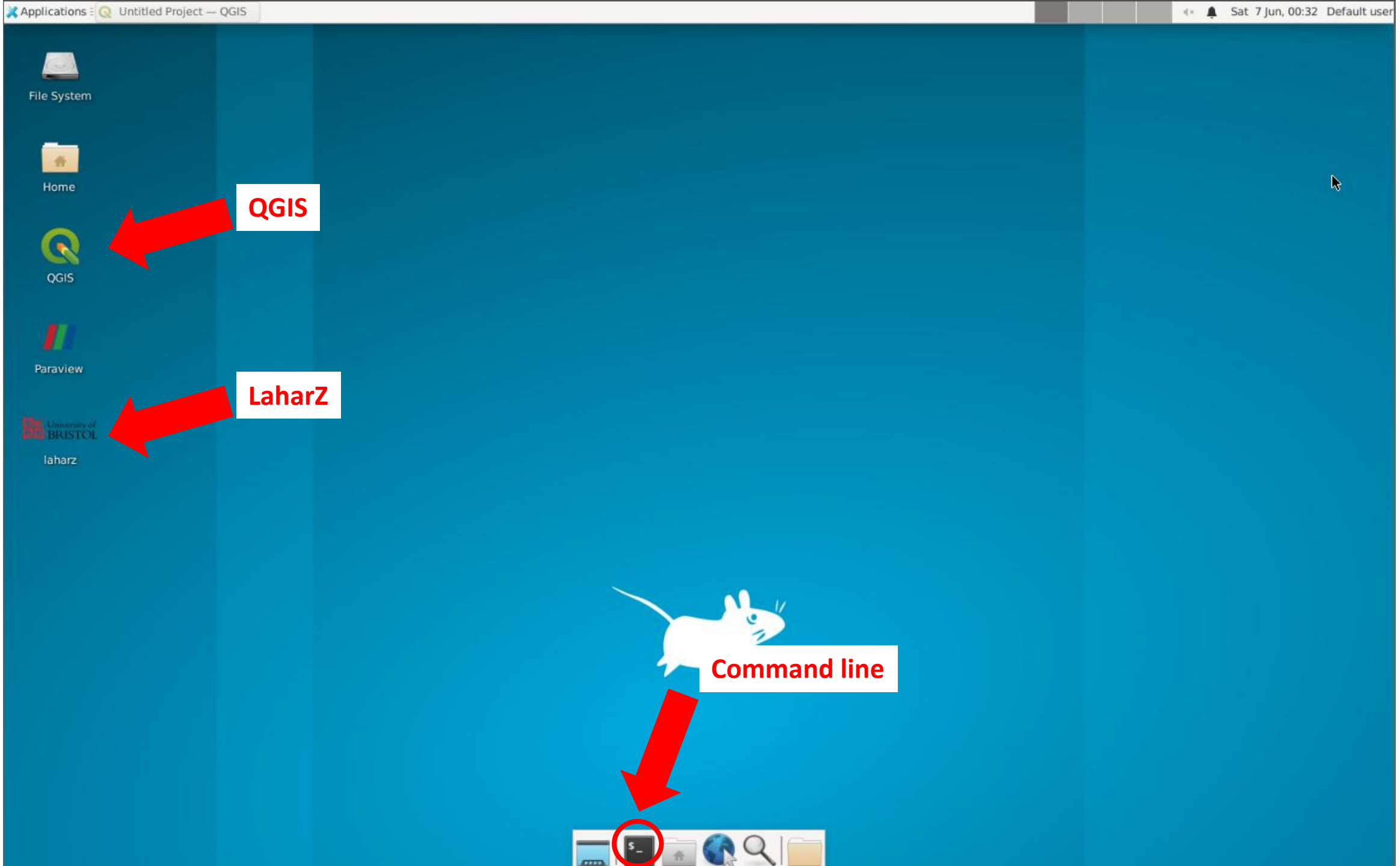


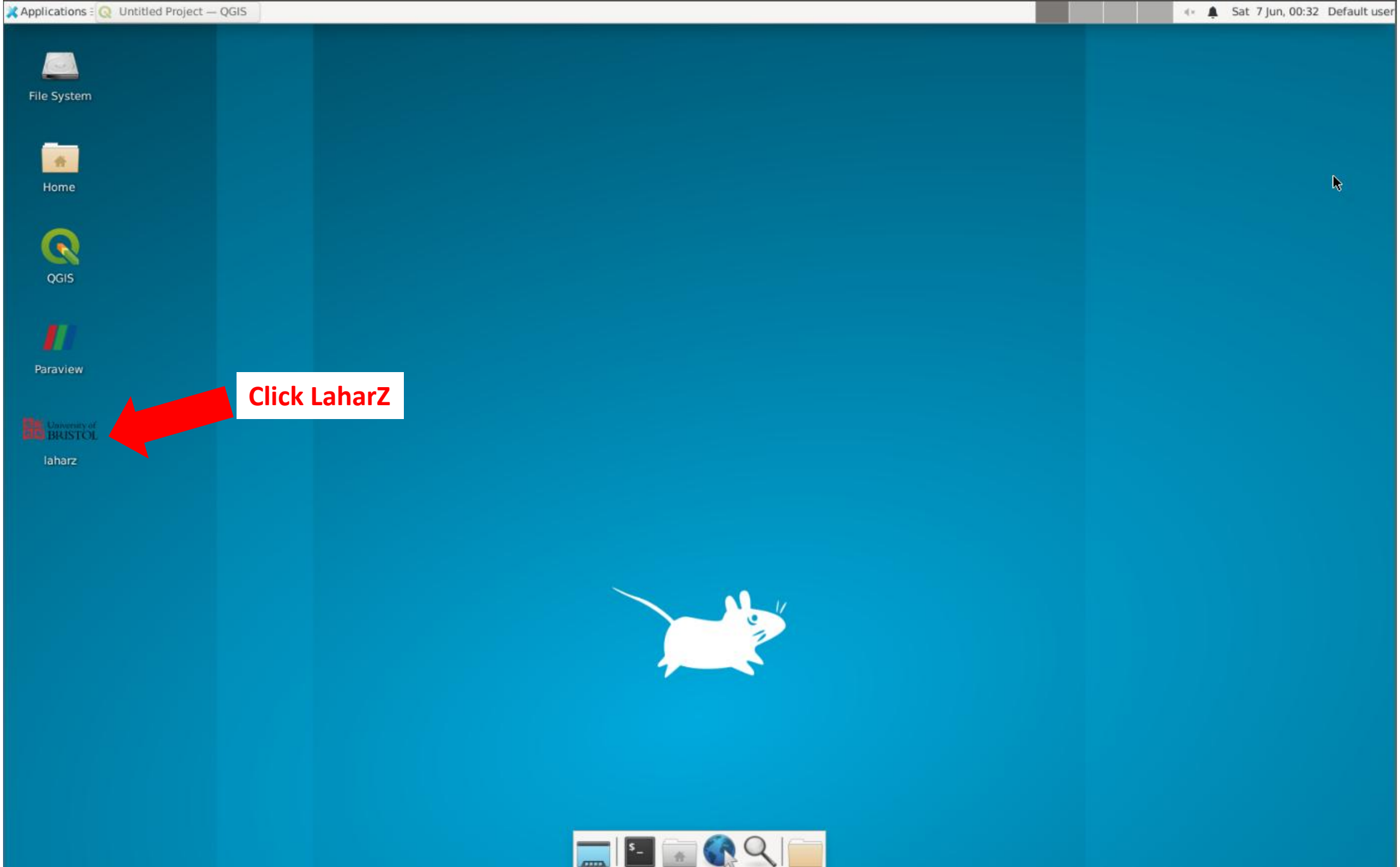
- We want to add the LaharZ model your home workspace
- Click the terminal
- Type: victor setup
- Hit enter on your keyboard
- Type: 10 for LaharZ
- Hit enter
- Close the terminal

- Check to see if the LaharZ folder was created in your space
- Click the home folder and look for the LaharZ folder (You will have different files than me, don't worry!)

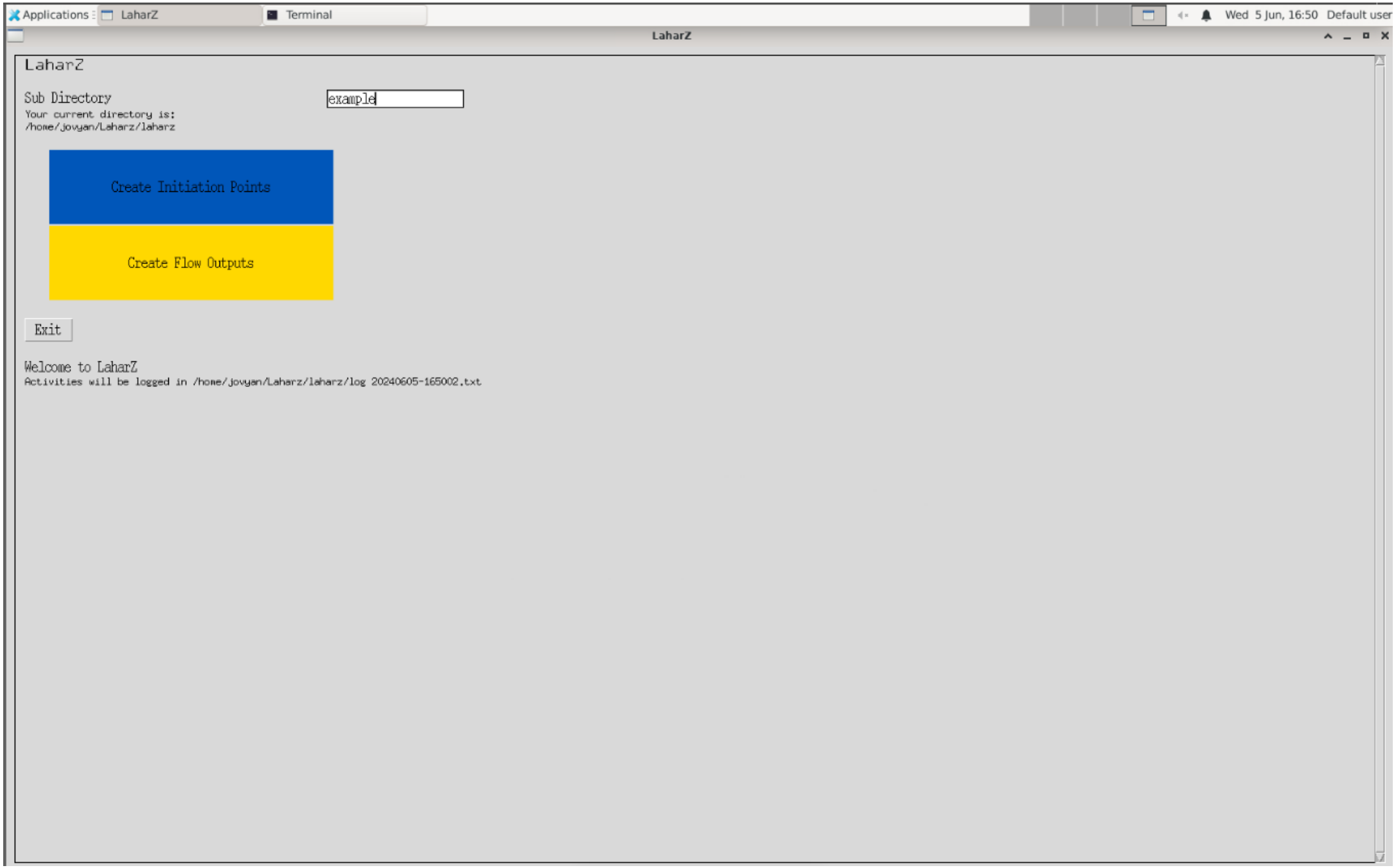



File Edit View Run Kernel Git Tabs Settings Help
Launcher
Filter files by name
/
Name
Modified
Conflow 3 months ago
Desktop 12 days ago
IMEX_LAVA 3 months ago
IMEX_SfloW_2D 2 months ago
Laharz 3 months ago
Molasses 3 months ago
mr_lava_loba 3 months ago
shared 23 seconds ago
shared-readwrite 23 seconds ago
Sulfur_X 3 months ago
Titan2D 2 months ago
Notebook
Click on the “Desktop” icon
Python [conda env:notebook] *
Desktop [↗]
panel [↗]
R [conda env:notebook] *
RStudio [↗]
Shiny [↗]
Console
Python [conda env:notebook] *
R [conda env:notebook] *
Other
Terminal
Text File
Markdown File
Python File
R File
Show Contextual Help
Simple
0
0
Mem: 510.47 / 8192.00 MB
Launcher
1


Applications
Untitled Project — QGIS
Sat 7 Jun, 00:32
Default user
File System
Home
QGIS
QGIS
Paraview
LaharZ
laharz
Command line


Applications
Untitled Project — QGIS
Sat 7 Jun, 00:32
Default user
File System
Home
QGIS
Paraview
Click LaharZ
laharz


Applications
LaharZ
Terminal
Wed 5 Jun, 16:50 Default user
LaharZ
LaharZ
University of
BRISTOL
Version: 2.0.0 stand alone


Applications
LaharZ
Terminal
Wed 5 Jun, 16:50 Default user
LaharZ
LaharZ
Sub Directory
example
Your current directory is:
/home/jovyan/Laharz/laharz
Create Initiation Points
Create Flow Outputs
Exit
Welcome to LaharZ
Activities will be logged in /home/jovyan/Laharz/laharz/log 20240605-165002.txt